Reference Event 2: The Arkema Crosby Fire, August 2017

Characterization Against the Chemical Sector Design Basis

DBA-CHEM-RE-2 — Reference Event Instrument, DBA-CHEM Series

Lineage: Issued under the Chemical Sector Framework v0.9; DBA-MA Parent Framework; conferred on the proposal of Chemical Production and Synthesis (DBA-CHEM-PR-FC1)

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Timothy E. Roxey

Eclectic Technologies

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the loss of temperature control and the fires of organic peroxides at the Arkema Inc. plant at Crosby, Texas, in the flooding that followed Hurricane Harvey: the loss, over 27 and 28 August 2017, of the plant's normal power, its emergency generators, its liquid nitrogen cooling and its means of moving product; the transfer of the product into nine refrigerated trailers; the evacuation of the crew and of residents within a mile and a half; the exposure of the responders enforcing that zone on a highway through it; and the burning of one trailer on 31 August, two on 1 September and the remaining six by controlled burn on 3 September. It reads the event back through the Chemical Sector Design Basis, as DBA-CHEM-PR-FC1 (*Chemical Production and Synthesis*) applies it to the fixed plant, and asks which provisions bore on it, how each performed, and what the class did and did not correctly draw from it.

The event is the class's empirical case for the loss of power and supporting utilities (§5.4), the event its organic peroxide reference plant was chosen to bound (§1.3), and the source of its required function that backup power and refrigeration for temperature-controlled inventory not share the primary's exposure to flood (§5.4).

1.2 Position and Scope

This is a reference event instrument in the sense of the series convention: it reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. It was issued under the Chemical Sector Framework v0.9, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 6 October 2026 on the proposal of *Chemical Production and Synthesis*. Its partner under the same class is DBA-CHEM-RE-1 (*TRITON Attack*).

On the author's decisions of 8 October 2026, the primary source is the Chemical Safety Board's final report, *Organic Peroxide Decomposition, Release, and Fire at Arkema Crosby Following Hurricane Harvey Flooding* (Report No. 2017-08-I-TX, May 2018), read in full; ¶n gives its paragraphs. The company's, Harris County's and the State of Texas's accounts and the prosecution of the company, two executives and the plant manager are recorded as their own, with no finding of fault, of recklessness or of the accuracy of any party's representations. The scope runs from landfall to the lifting of the evacuation on 4 September, with the Board's recommendations and the prosecutions as context. Times are Central Daylight Time.

The primary sector is Chemical. The plant, its inventory, its warehouses and the trailers used as storage on plant property lie within the production class (§1.2, §8.3). The officers and medical personnel exposed were Harris County's and its emergency services', and the route on which they were exposed, U.S. Highway 90, was held open for the wider Harvey response, so

Emergency Services and Transportation are proposed as impacted sectors. The constructs applied are cited, not restated: FD-BL (*The Bright Line*), FD-BR (*The Broker*), FD-LD (*Layer Decomposition and Determinism*), FD-LS (*Local Sovereignty*), FD-BL-D1 (*Mode of Discharge*) and FD-BL-D2 (*Indication Integrity*); FD-REG holds their versions. Section references of the bare form §N are to *Chemical Production and Synthesis* v0.3, the version this instrument read.

2. Sources and Attribution

2.1 Sources and Their Limit

The principal source is the Board's final report, read in full (paragraphs 1 to 352 and Appendices A to C), with its executive summary, news release and recommendation status-change summaries. The others are the Texas Commission on Environmental Quality's draft report of 3 October 2017, the joint statement of the Environmental Protection Agency and the Commission of 3 September 2017, the National Hurricane Center's report on Harvey, the company's statements, Senator Thomas Carper's letter of 1 September 2017 and the reporting cited in the predecessor. No non-public material is used. The risk management plan, the indictments and the court's rulings have not been read in their originals.

2.2 Foreseeability, Fault and the Contested Points

On what the plant should have foreseen, the accounts differ. The Board found that the plant was built before any flood map of its area; that a revision of 2007 placed the low-temperature warehouses and the electric supply in the 100-year floodplain, which the plant did not take into its process safety management (¶154, ¶156, ¶251); that it had no single flood design basis (¶166); that its process hazard analysis did not assess flooding; that its safeguards could likely have protected it in a 100-year flood (¶31, ¶350); and that the rainfall had less than a one-in-one-thousand-year likelihood (¶338). The company called the floodwaters unprecedented. The prosecutions, for reckless emission and for assault on a charge that the company's representations put deputies in harm's way, ended in dismissals and directed verdicts in 2020, with no conviction; the company's position was that Harvey was an unforeseeable act of nature. All of these are their sources', and this instrument makes no finding between them.

Ten smaller differences of count, depth and time are stated in the predecessor; this instrument uses the Board's figures. The record is closed in its principal parts; OI-2 records what remains open.

3. The Event

3.1 The Plant and Its Inventory

The plant made about thirty organic peroxide products, which decompose at a rate set by temperature, and served as a distribution hub (¶81). It held about 367,000 pounds of refrigerated

product, eleven products whose self-accelerating decomposition temperatures ran from 23 to 113 °F, in seven low-temperature warehouses.

Four layers stood between that inventory and its decomposition, on the Board's account: at least two electric refrigeration systems per warehouse; seven diesel emergency generators, one with its base about two feet above the ground; a liquid nitrogen cooling system independent of electricity; and eleven refrigerated trailers, the last backup, each able to hold its temperature for more than a week on a full tank of diesel carried beneath its refrigeration unit. The base flood elevation at the site is 52 feet (¶159), the laydown area at the north end, the plant's high ground, stands at 52 feet, and the difference between the 100-year and 500-year flood elevations at the site is less than a foot (Figure 53). The company's risk management plan covered sulfur dioxide and isobutylene, whose tanks drove its worst-case scenario, reported as reaching 1.1 million residents over 23 miles; the company did not consider the low-temperature warehouses covered by that plan or by the Process Safety Management standard (¶129, ¶231, ¶235).

3.2 The Storm

Harvey made landfall at 22:00 on 25 August and stalled, in what the National Hurricane Center calls the most significant tropical cyclone rainfall event in United States history. The plant's gauge recorded more than 39 inches, and the Board puts the water more than two feet above the 500-year flood elevation at the site; the staff had expected one to two feet.

3.3 Preparation and the Loss of the Layers

The plant halted production on 25 August and activated its ride-out crew (¶64, ¶68, ¶70). As the water rose the crew cut each warehouse's power ahead of it and moved the product into trailers at the laydown area. At about 05:00 on 28 August the last generator was shut down; the liquid nitrogen piping was under water; floodwater disabled the forklifts and semi-tractors, so the trailers could no longer be moved; and empty trailers failed as their fuel tanks filled. More than 350,000 pounds of product ended in nine trailers, three stranded in floodwater by the warehouses. The corporate crisis team considered moving the trailers off site and concluded that neither drivers nor vehicles could reach it; the Board observes that a move off site would have needed a decision much earlier, possibly before landfall, because drivers licensed for hazardous materials could take days or weeks to find. The crisis team told Harris County's responders that if the trailers failed the product would begin to burn within a few days.

3.4 Evacuation and the Zone

The crew was taken out by boat on 29 August. Telemetry on six of the nine trailers gave the air temperature and whether refrigeration ran; it measured air, not product. The zone was set at a mile and a half on modeling of the trailers' combustion with releases from the sulfur dioxide and isobutylene tanks, a sulfur dioxide release that could have reached several miles being judged not credible (¶110, ¶286 to ¶290, ¶295). About 205 residents were evacuated. With Interstate 10

flooded, Highway 90, running through the zone past the plant, was the only viable route for the wider response and was left open to eastbound traffic (¶17, ¶18).

3.5 The Exposures on Highway 90

Just before midnight on 30 August two officers on Highway 90 near the plant drove into what they reported as a “large cloud of gas” and reported irritation. Telemetry showed no temperature spike, and at 00:30 officials judged the cloud weather and reopened the highway; the Board states that the trailer that likely produced it had no telemetry, which was not known at the time. More officers and medical personnel then drove through clouds, and the highway was closed until the all-clear. In all, 21 emergency responders reported to the hospital (¶124).

3.6 The Fires and the Controlled Burn

One trailer burned at about 02:00 on 31 August and two at about 17:00 on 1 September. The unified command and the company then decided on a controlled burn, sequenced so that the last trailer would shield an isobutylene tank about forty yards from the nearest (¶134, ¶137), and burned the six in calm winds on 3 September (¶140). The zone was lifted on 4 September (¶141, ¶142).

3.7 What Was Lost

The refrigerated inventory, more than 350,000 pounds, was lost entirely, with nine trailers. No death is recorded. Twenty-one emergency responders sought medical attention, about 205 residents were evacuated for about a week, and a regional highway carrying the Harvey response was closed for four days. Some 23,000 pounds of organic material reached the floodwater from the wastewater tanks.

3.8 Outcome

The Board’s five recommendations, to the plant, the company, the Center for Chemical Process Safety and Harris County, are all closed as acceptable action, the plant now meeting the county’s revised 500-year flood elevation standard; it reiterated to the Environmental Protection Agency Recommendation 2001-1-H-R3, that the Risk Management Program cover catastrophic reactive hazards (§15.1). The criminal case ended on 1 October 2020 with no conviction.

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-3), which made the loss possible, and effect elements (E-1 to E-4), which carried it to the people around the plant and to its end, as Table 4-1 sets out. Composed, they describe a plant whose four layers of protection for one temperature-controlled inventory all stood within reach of one flood; a margin of days spent while the decision that could have used it had already passed; a plant abandoned by its crew and watched in part; and a release managed by off-site authorities from inside its footprint, ending in

a burn they chose. That is the class's chain R, release (§3.2), by its loss of power and supporting utilities (§5.4), brought on by natural-hazard exceedance (§5.7), at its organic peroxide reference plant (§1.3).

Table 4-1. Event elements and their mapping. Section references are to Chemical Production and Synthesis v0.3 unless another instrument is named.

Ref	Description	Provisions engaged	Finding
I-1	Flooding over five feet, more than two feet above the 500-year elevation, at a plant part in the 100-year and part in the 500-year floodplain, built before any flood map, with no single flood design basis; the Board found its safeguards adequate for a 100-year flood	§3.3, §5.7, §8.4, §10.1	F-3, F-4
I-2	Primary and backup refrigeration, generators, liquid nitrogen piping, trailer fuel and refrigeration, forklifts and tractors all exposed to one flood	§4.2, §5.4, §5.7	F-1, F-2
I-3	More than 350,000 lb of organic peroxide, SADT 19–70 °F in the trailers; days of margin on the company's estimate	§1.3, §3.3, §5.4	F-2, F-9
E-1	Three trailers stranded on low ground short of the laydown area; no move off site possible; the Board places the decision before landfall	§1.2, §5.4, §6.7, §8.3	F-2
E-2	Crew evacuated; telemetry on six of nine trailers, air not product; likely source of the first cloud unmonitored	§5.3, §6.7, §7.2	F-7
E-3	1.5-mile zone on modeling of the trailer fire with SO2 and isobutylene releases judged not credible beyond it; Highway 90 open eastbound through it; 21 responders evaluated	§4.2, §4.4, §4.5, §6.3	F-5, F-6
E-4	Fires on 31 August (1 trailer) and 1 September (2); controlled burn of 6 on 3 September, the last sequenced to shield the isobutylene tank	§4.2, §6.7, §6.8	F-5, F-8

5. Tier Determination

The four military-action characteristics are read as *Chemical Production and Synthesis* §6.2 states them for the production plant, against the event as the record establishes it, with no finding of fault.

5.1 Intentionality

Absent. The initiating event was a hurricane, and no charge alleged an intent to release.

5.2 Persistence

Absent as an adversary's. The hazard itself persisted, rain and then standing water over about ten days, and that duration set the event's course.

5.3 Precision

Absent. The flood took what stood low, in the order of the ground. Nothing was chosen.

5.4 Denial

Present in effect, without an actor. The flood denied power and its backups, the means of moving the inventory, road access, the crew and a full view of the inventory; what the class says an adversary would arrange, the act timed to a concurrent loss of power and response (§6.2), the storm supplied.

5.5 Determination

The event does not meet the threshold for the military-action extension. Intentionality, persistence and precision are absent; denial is present in effect, by the hazard and not by any actor. It is a natural-hazard exceedance at a production plant, flooding beyond the plant's design basis removing a utility and its backups together, in the class's natural-hazard exceedance and loss-of-utility categories (§5.7, §5.4), chain R, at the organic peroxide reference plant (§1.3), the military-action extension not engaged. The provisional placement is confirmed. That is the tier determination this instrument returns for the DBA Reference Event Register.

5.6 The Scenario Realized

The event reached, by flood, the class's second declared residual, sustained denial of utilities and site access outlasting the plant's hold on its inventory (§6.8). What a directed actor would do through the path the event exposed, choosing the layer, the moment and what burned first, runs forward and is offered to Project 9 as use-case material. Its bearing here is recorded in F-2, F-5 and F-7.

6. Characterization Against the Design Basis

Section references are to *Chemical Production and Synthesis* v0.3, the version this instrument read. The class has since taken in R-1 to R-9 at its v0.4.

6.1 §5.4 and §5.7 Backup Independent of the Common Cause

Held, with a gap. The Board's central finding, a common mode of failure in every layer, is the class's requirement read backward; the event adds that the equipment and people that move inventory to its backup are themselves a layer. Finding F-1.

6.2 §1.3, §3.3, §5.4 and §6.7 The Grace Period and When It Was Spent

Missed. The margin was days, not hours, and was spent waiting: the decision that mattered fell between forecast and flood, for which the class has no parameter. Finding F-2.

6.3 §3.3, §5.7, §8.4 and §10.1 The Design-Basis Flood

Missed. The class requires layers to remain within their design basis under an exceedance but states no hazard against which the plant is designed, and the plant had none. Finding F-3.

6.4 §3.3, §4.3, §4.4, §8.4 and §10.1 The Plant's Worst Case and the Event

Held, with a gap. The class leaves its parameters to be checked against the analyses the plant holds; here they omitted the inventory the class chose the plant for, and the decomposition path has no toxic endpoint. Finding F-4.

6.5 §4.2, §5.2 and §6.3 What the Fire Could Reach

Held, with a gap. The trailers were lost to one cause, as the class counts; the event adds escalation to adjacent inventory and a source term arriving in stages. Finding F-5.

6.6 §4.5 and §6.3 Who Stood in the Footprint

Missed. The people harmed were those enforcing the zone, on a route held open through it, whom the exposure stage does not count. Finding F-6.

6.7 §5.3, §6.7 and §7.2 What Could Be Seen of the Abandoned Plant

Missed. The class does not provide for a plant whose crew is gone and whose inventory is known only remotely. At Crosby the telemetry covered six of nine trailers, measured air rather than product, and the highway was reopened on a reading that did not cover the likely source; the indication's absence was not shown as absence. The company's representations about its monitoring were the subject of the assault charge, which was dismissed, and this instrument makes no finding on them. The requirement is physical, not cyber. The company's corporate crisis team held that information and passed it to the authorities, which is the coordination class's, *Chemical Enterprise Coordination and Control*. Finding F-7.

6.8 §6.7 and §6.8 The Residual Reached, and the Burn That Ended It

Held, with a gap. A flood reached the residual the class declares for sustained denial; the end state was a controlled burn decided off site, and the crew was withdrawn under the same hazard. Finding F-8.

6.9 §1.3 and §5.4 The Class's Account of the Event

Held, with a gap. Accurate in substance; the fires, the trailers, the layers lost, the twenty-one and "some hours later" needed qualifying. Finding F-9.

6.10 §3.2 and §5.5 Chain S

Not exercised. Nothing in the record shows that a dependent facility lost supply.

6.11 §5.3, §6.4 and §7.3 Cyber and the AI-Governance Design Basis

Not exercised. No compromise, and no component within the scope of *Layer Decomposition and Determinism* §4.3, bore on the event; the physical indication requirement is read at Section 6.7.

6.12 §2.3 The Cluster

Not exercised. Harvey struck the Gulf Coast cluster, and the Board records other releases across it (¶313 to ¶319); this instrument reads one plant.

6.13 Summary of Design Basis Performance

Of the twelve readings, backup independence, the worst case, the fire's reach, the residual and the class's account Held, with a gap; the grace period, the design-basis flood, the footprint and the abandoned plant's indication were Missed; and chain S, cyber and the AI-governance design basis, and the cluster were Not exercised.

7. Findings

Section references are to *Chemical Production and Synthesis* v0.3 unless another instrument is named.

7.1 F-1 — One Flood Took Every Layer

Redundant refrigeration, emergency generators, liquid nitrogen cooling and refrigerated trailers all failed to the same flood, the Board finding a common mode of failure in every layer of protection. The finding confirms the class's required function that backup power and refrigeration for temperature-controlled inventory not share the primary's exposure to flood (§5.4), its attention to the common-cause loss of a utility and its backup (§5.7), and its refusal to credit mitigation that the event defeats (§4.2), and records that the equipment and people that move inventory to its backup, forklifts, tractors and licensed drivers, are a layer to which the same rule applies.

Affected: §1.2, §4.2, §5.4, §5.7. Disposition: R-1.

7.2 F-2 — The Grace Period Was Days, and the Decision Fell Before the Flood

The first fire came about 69 hours after the last warehouse lost power, and the trailers could hold for more than a week; the move off site that could have used the margin needed, on the Board's account, a decision possibly before landfall, and by the time the utilities were lost the roads, vehicles and drivers were lost with them. The finding is that the class should qualify "some hours later" (§1.3) to state the grace period of the warehouse and of its last credited backup separately, more than four hours for the first on the plant's own analysis and days for the second at Crosby; that it should add, beside the grace period, a decision margin running from a credible forecast of the hazard to its arrival, within which inventory is reduced, moved or raised, with the trigger stated in advance; that a safe state reached by moving inventory is credited only if the means of moving it, on site and off, are protected against the cause, and that staging it on site above the design-basis hazard during preparation, which the Board found would likely have held the trailers (¶171 n., ¶349), is among the moves the margin covers; and that the hazardous-materials carriers and drivers on which a move off site depends are noted to Bulk Chemical Transport.

Affected: §1.3, §3.3, §5.4, §6.7; Bulk Chemical Transport. Disposition: R-2.

7.3 F-3 — The Class Names No Design-Basis Hazard

The plant was built before any flood map, mapped in 2007 into the 100-year and 500-year floodplains, warned by its then insurer in 2016, and analyzed without flooding, and the Board found no single flood design basis for it (¶166); the Board found its safeguards adequate for a 100-year flood, and Harvey's water stood more than two feet above the 500-year elevation. The class requires layers to remain within their design basis under an exceedance (§5.7) but states no hazard parameter (§3.3). The finding is that the class should state a design-basis natural hazard as a parameter, for flood no lower than the most severe mapped floodplain, recording the map and its date, held in process safety information and renewed with the process hazard analysis cycle as the Board recommended; that it should state what the plant must do when the hazard exceeds that basis, the margin to the point at which every layer is lost and the declared residual beyond it; and that the framework, whose event categories are still stubs and do not yet name natural-hazard exceedance (framework §4.2), take up the category the class developed at §5.7.

Affected: §3.3, §5.7, §8.4, §10.1; framework §4.2. Disposition: R-3.

7.4 F-4 — The Plant's Worst Case Was Not the Event

The plant's risk management plan covered only its sulfur dioxide and isobutylene, whose release drove its worst case, and the organic peroxide warehouses were outside it and outside the Process Safety Management coverage (¶129, ¶231, ¶235); the event was an organic peroxide decomposition and fire, the Board found that had reactives been covered the company may have identified its flood vulnerability (¶248), and it reiterated Recommendation 2001-1-H-R3 that the Risk Management Program cover catastrophic reactive hazards (§15.1). The class leaves its parameters to be validated against such analyses (§3.3, §10.1) and requires consistency with

them (§8.4). The finding is that the class should state that, where a plant holds reactive or temperature-controlled inventory, its source term and footprint are set from the class's own governing event for that inventory, whether or not the plant's regulatory analyses include it; and that the release model should give the decomposition path, which carries a thermal endpoint (§4.4), a toxic endpoint for its smoke and decomposition products, which no safety data sheet covered and the dispersion models did not anticipate (¶291, ¶294), with detection at the perimeter that can tell such a cloud from weather.

Affected: §3.3, §4.3, §4.4, §8.4, §10.1. Disposition: R-4.

7.5 F-5 — The Fire Reaches the Next Inventory

The zone was set on modeling of the trailer fire with releases from the plant's sulfur dioxide and isobutylene tanks, a sulfur dioxide release that could have reached several miles being judged not credible (¶286 to ¶290); the isobutylene tank stood about forty yards from the nearest trailer and the controlled burn was planned so that the last trailer to burn would shield it (¶134, ¶137); the trailers failed over three days, not together. The finding is that the source term should count, beside vessels sharing a cause (§4.2), inventories within reach of another's fire or decomposition; that the siting requirement for storage (§5.2) should separate temperature-controlled reactive inventory from toxic and flammable inventory, or count them as one source term; and that the footprint and the duration of a zone should account for a source term that arrives in stages.

Affected: §4.2, §5.2, §6.3. Disposition: R-5.

7.6 F-6 — The Zone Was Enforced From a Road Through It

Highway 90 ran through the zone as the only viable route for the Harvey response, was reopened on an indication that did not cover the source, and the officers and medical personnel enforcing the zone were exposed on it, 21 responders reporting to the hospital; the Board found it should have been closed once a report showed decomposition might be occurring (¶35, ¶299) and recommended protection for those enforcing perimeters and for personnel moved through the zone. The finding is that the exposure stage (§4.5) should count transport routes through the footprint and the responders who must move through it, and that the emergency plan should state, in advance, the closure of such routes and the protection of perimeter personnel against the decomposition path; and that the event is noted to the emergency services series and the transportation series, whose decisions on how they count it are their own.

Affected: §4.5, §6.3; DBA-ESS; DBA-TRAN; DBA Reference Event Register. Disposition: R-6, R-10.

7.7 F-7 — The Abandoned Plant Was Seen in Part

After the crew's evacuation the inventory was watched by telemetry on six of nine trailers, measuring air, not product; the trailer that likely produced the first cloud had none, and the highway was reopened on a reading that did not cover it. The finding is that the class should require, for temperature-controlled inventory, remote indication of the state of every container, of product temperature where it can be had, powered independently of the flood-exposed supply, with its coverage stated and any gap shown as a gap to whoever acts on it, and that the passing of that indication from the company to the emergency authorities is a coordination function of Chemical Enterprise Coordination and Control.

Affected: §5.3, §6.7, §7.2; Chemical Enterprise Coordination and Control. Disposition: R-7.

7.8 F-8 — A Flood Reached the Residual the Class Declares for War

The class's second declared residual, sustained denial of utilities and access outlasting the plant's hold on its inventory, with evacuation as the response (§6.8), was reached by a natural hazard. The end state was a controlled burn decided and carried out by the unified command with the company, after the crew had been withdrawn under the same hazard. The finding confirms §6.8 and is that the class should record that the residual is reachable in the non-adversarial domain at a plant that does not meet Section 5; that the end state of §6.7 may be reached by deliberate controlled destruction, and should state what that needs, favorable meteorology, safe entry, sequencing against adjacent inventory, and the off-site authority that decides it; and that the endurance of the small crew (§6.7) is bounded by the crew's own safety under the same hazard, the Board noting that even trailers staged on high ground would have needed the crew on site to monitor them (¶171 n.).

Affected: §6.7, §6.8. Disposition: R-8.

7.9 F-9 — Withdrawn at v0.2

Its substance qualified the class's own account of the event at *Chemical Production and Synthesis* §1.3, §5.4, §8.4, §9.2 and §10.7: three fires, on 31 August, on 1 September and by controlled burn on 3 September, of nine refrigerated trailers; four layers lost, with the means of moving the inventory; about 205 residents evacuated and more than 200 unable to return for a week; the twenty-one as emergency responders exposed on a highway held open through the zone; "some hours later" qualified to separate the warehouse's grace period from the backup's, as F-2 sets out; the Center for Chemical Process Safety's monograph *Assessment of and Planning for Natural Hazards* named beside the Board's recommendations; and the event recorded as conferred. That is a correction to another instrument rather than a test of the design basis. The class took it in at its v0.4, so no erratum is owed. R-9 is withdrawn with it. Its register element, Emergency Services and Transportation as impacted sectors, stands as R-10.

8. Recommendation Register

Priority is stated on the series' one-to-three scale, where 1 is the most operationally consequential. R-1 to R-8 dispose to *Chemical Production and Synthesis*, which took them in at its v0.4; the carrier element of R-2 to *Bulk Chemical Transport*, which took it in at its v0.4; the framework element of R-3 to the Chemical Sector Framework, which took it in at its v0.10; the coordination element of R-7 to *Chemical Enterprise Coordination and Control*, which took it in at its v0.2; the emergency-services and transportation elements of R-6 as notes to those series; and R-10 to the register.

ID	Recommendation	Affected	Direction	Pri
R-1	Confirm the independence rule	§1.2, §4.2, §5.4, §5.7	Confirm that backup power and refrigeration must not share the primary's exposure to a common cause, on the Board's finding of one common mode across four layers; extend the rule to the equipment and people that move inventory to its backup.	3
R-2	Set the decision margin	§1.3, §3.3, §5.4, §6.7; Bulk Chemical Transport	State the grace period of the warehouse and of its last credited backup separately, in days where it is days; add a decision margin from credible forecast to arrival, with a stated trigger to reduce, move away or stage on high ground; credit a move only if its means are protected against the cause; note the dependence on hazardous-materials carriers to Bulk Chemical Transport.	1
R-3	Name the design-basis hazard	§3.3, §5.7, §8.4, §10.1; framework §4.2	Add a design-basis natural hazard parameter, for flood no lower than the most severe mapped floodplain, map and date recorded, held in process safety information and renewed each hazard analysis cycle; state the margin to loss of every layer and the declared residual beyond it; ask the framework, whose stub categories do not yet name natural-hazard exceedance, to take up the category developed at §5.7.	1
R-4	Set the reactive source term from the class	§3.3, §4.3, §4.4, §8.4, §10.1	Set source term and footprint for reactive and temperature-controlled inventory from the class's governing event whether or not regulatory analyses include it; give the decomposition path a toxic endpoint for its smoke and decomposition products, and perimeter detection that distinguishes it from weather.	2
R-5	Count escalation between inventories	§4.2, §5.2, §6.3	Count in the source term inventories within reach of another's fire or decomposition; separate reactive from toxic and flammable storage or count them together; size footprint and zone duration for a staged source term.	2

R-6	Count routes and responders in the footprint	§4.5, §6.3; DBA-ESS; DBA-TRAN	Count transport routes through the footprint and the responders who move through it; state route closure and perimeter protection in the emergency plan in advance; note the event to the emergency services and transportation series, whose counting is their own.	1
R-7	Indicate the abandoned inventory	§5.3, §6.7, §7.2; Chemical Enterprise Coordination and Control	Require independent remote indication of every temperature-controlled container, product temperature where possible, coverage stated and gaps shown as gaps; state the passing of it to emergency authorities as a coordination function.	2
R-8	Record the residual and the controlled burn	§6.7, §6.8	Record that the second residual is reachable by natural hazard at a plant not meeting Section 5; recognize controlled destruction as a route to the §6.7 end state and state what it needs and who decides it; bound crew endurance by the crew's safety under the same hazard.	2
R-9	Withdrawn at v0.2	§1.3, §5.4, §8.4, §9.2, §10.7	Its substance, qualifying the class's account, was taken in by Chemical Production and Synthesis at v0.4; withdrawn with F-9.	—
R-10	Name the sectors	DBA Reference Event Register	Ask that the register name Emergency Services, for the responders exposed, and Transportation, for Highway 90, as impacted sectors.	3

9. Open Items

OI-1, the register row, is closed: Project 0 entered the tier determination of Section 5.5 with Emergency Services and Transportation as impacted sectors. OI-2 stands: the plant's risk management plan of 2014, the indictments and the court's rulings in their originals, and the outcome of the responders' and residents' civil suits; any would bear on Sections 2 and 3 and on F-4. OI-3 is closed: *Chemical Production and Synthesis* took in R-1 to R-9 at its v0.4 and records the event as conferred and written.

10. Standing of This Document

10.1 What This Instrument Establishes

This instrument establishes where the Arkema Crosby fire sits within the Chemical analytical structure and what *Chemical Production and Synthesis* did and did not correctly draw from it. It modifies neither the class nor the framework, and makes no finding of fault, of recklessness, of the accuracy of any party's representations, or of law. It should be revised on any class revision that dispositions these findings, or on the material of OI-2.

10.2 Relationship to the Other Production-Class Event

The TRITON attack (DBA-CHEM-RE-1) is the deliberate case, an adversary who reached the safety layer first and was stopped by an error in its own code; Crosby is the natural case, every layer lost to one flood. They are the two ways to the shape of failure the class takes from Bhopal, every protective layer unavailable at once (§2.4). Crosby shows how long the margin lasted once the layers were gone.

10.3 Relationship to the Series' Other Instruments

Graniteville (DBA-CHEM-RE-3) and East Palestine (DBA-CHEM-RE-4) are releases in which the zone also decided who was exposed, and F-6 is made with them in view.

References

The sources are those named in Section 2.1 and the reporting cited in the predecessor, where the full list with titles and dates is kept. DBA instruments: the Chemical Sector Framework v0.9; *Chemical Production and Synthesis*, read at v0.3; *Bulk Chemical Transport*; *Chemical Enterprise Coordination and Control*; DBA-CHEM-RE-1, RE-3 and RE-4; the Foundational Definitions named in Section 1.2, with FD-REG; and the DBA-MA Parent Framework.

Revision History

Version 0.1, October 2026: issued in Project 9 under the Chemical Sector Framework v0.9, for the event conferred as DBA-CHEM-RE-2 on the proposal of *Chemical Production and Synthesis*, on the author's decisions of 8 October 2026. Tier determination as Section 5.5 states it. Findings F-1 to F-9; recommendations R-1 to R-10.

Version 0.2, 10 October 2026: migrated to Project 11 under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), compressed under the Reference Event Compression Standard v0.3 and brought to house style. Cut, with every section keeping its number: the general description of the instrument; the sourcing sections, merged into Section 2 and OI-2; the narrative, to the facts the findings rest on; the decomposition's subsections, left to Table 4-1; the constructed analog of Section 5.6, offered to Project 9 as use-case material; Section 6, to the standard's verdicts, Section 6.7 kept fuller because *Chemical Enterprise Coordination and Control* cites it. the references, condensed. The triage proposed compressing F-1 to a confirmation and folding F-6 into F-5 with F-6 as a stub; F-1 is already a confirmation, and F-6 is cited separately by *Chemical Production and Synthesis* and DBA-CHEM-UC-1, so both are kept whole. Stubs: F-9 and R-9, whose qualification of *Chemical Production and Synthesis* §1.3, §5.4, §8.4, §9.2 and §10.7 the class took in at v0.4, so no erratum is owed. Unchanged: the designator, title, tier determination, statements of fault and foreseeability, sources, F-1 to F-8, R-1 to R-8 and R-10. OI-1 and OI-3 closed. Verified: every finding and recommendation number present at v0.1 is present here as an entry or a stub.